Good morning! My name is Keah Tandon, and I am a part-time student in the Educational Psychology and Research program, focusing on educational research. I am here today to present a collaborative project between Dr. Seaman and me, edsamplr: Simulating Data for Statistics Instruction.

SLIDE 2: Introduction

Many graduate students take introductory statistics courses in order to get training in quantitative data analysis. These courses ideally include data that comes from real research, information on the conditions for valid inference for the covered statistical tests, additional information on the consequences of violating those conditions, and practice working with a variety of distributional shapes.

The Problem

However, this ideal is often not achieved. Finding data is difficult and time consuming because one must search through literature for various statistical concepts that aren't the focus of the studies. The data is not always available or may not be in a useable format. Sometimes one must email the author and wait for a response. Additionally, multiple data sets are needed between introducing concepts, data for student practice, and more data for student assessment.

SLIDE 3: Features

We have a proposed solution to this problem, our R package: edsamplr. This package has functions that will simulate data to cover all the major concepts in an introductory statistics course, including binary, categorical, quantitative, matched pairs, and correlated data in addition to a wrapper function for many named distributions. Our functions all have an optional argument as well to generate a table of summary statistics since each sample is generated based on specified parameters.

Technical Details

To use this package, you call the appropriate function, input the proportion for binary or categorical samples, specified first four moments for a quantitative sample, and desired correlation or slope for correlated quantitative data. The quantitative functions use Headrick’s fifth-order polynomial method to generate data with valid probability density functions.

SLIDE 4

We are finalizing the first version of the package in the near future with plans to make it publicly available through GitHub and submitting it to be included in the CRAN. We also have a roadmap that includes unbalanced designs and additional functions for data and concepts that would be covered in a second-semester statistics course and maybe on into an experimental design course.

Here is a brief example demonstrating the use of our function. For an instructor teaching students about comparing two means, they might look to Rosenthal and Jacobson's Pygmalion study. As this data is not easily found, using the specified gain scores as parameters for sampling will provide students a data set and context to go with it.

SLIDE 5

Thank you! What questions do you have?